

TrES-3b

R-1321

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_220316 · created 2026-07-31T19:36:17+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459654.9102 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.016
Transit depth (Rp/Rs)^2	3.83 +/- 0.63 [%]
Orbital Inclination (inc)	82.53 +/- 0.74
Airmass coefficient 1 (a1)	0.954 +/- 0.036
Airmass coefficient 2 (a2)	0.033 +/- 0.013
Scatter in the residuals of the lightcurve fit is	3.73 %
Best Comparison Star	#3 - [494, 325]
Optimal Aperture	3.62
Optimal Annulus	15.99
Transit Duration (day)	0.0636 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.016 vs 0.16309 (deviation 1.69σ)
Duration fit vs published	0.0636 d vs 0.0567 d (deviation 0.97σ)
Mid-time O–C	-5.0 min
Depth SNR	10.1
Residual scatter	3.73 %

Light curve

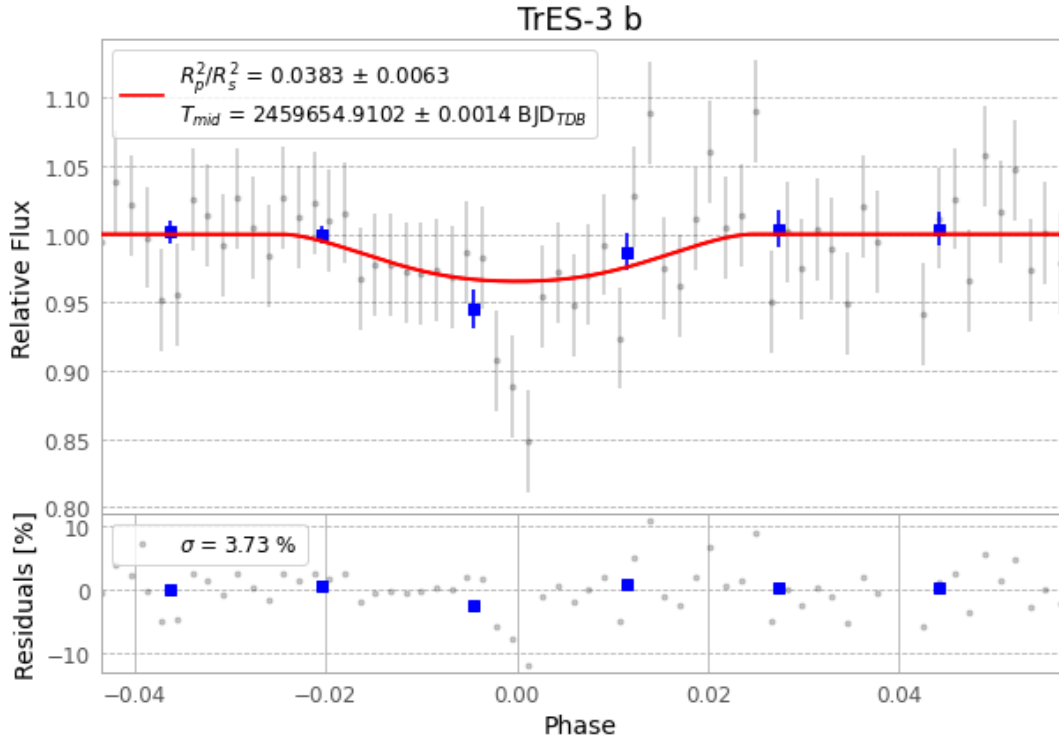


Figure 1 — FinalLightCurve_TrES-3 b_2022-03-16.png (SHA-256 d991544f10c26790...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [195, 290], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4a3a9caeba58f99c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), TRES-3220316082712.FITS → TRES-3220316113612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-220316.log (69109d34e664...), FinalLightCurve_TrES-3 b_2022-03-16.png (d991544f10c2...), FinalLightCurve_TrES-3 b_2022-03-16.csv (7b1fda4f324b...)

Compute resources

Wall time 245.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 19:36Z.